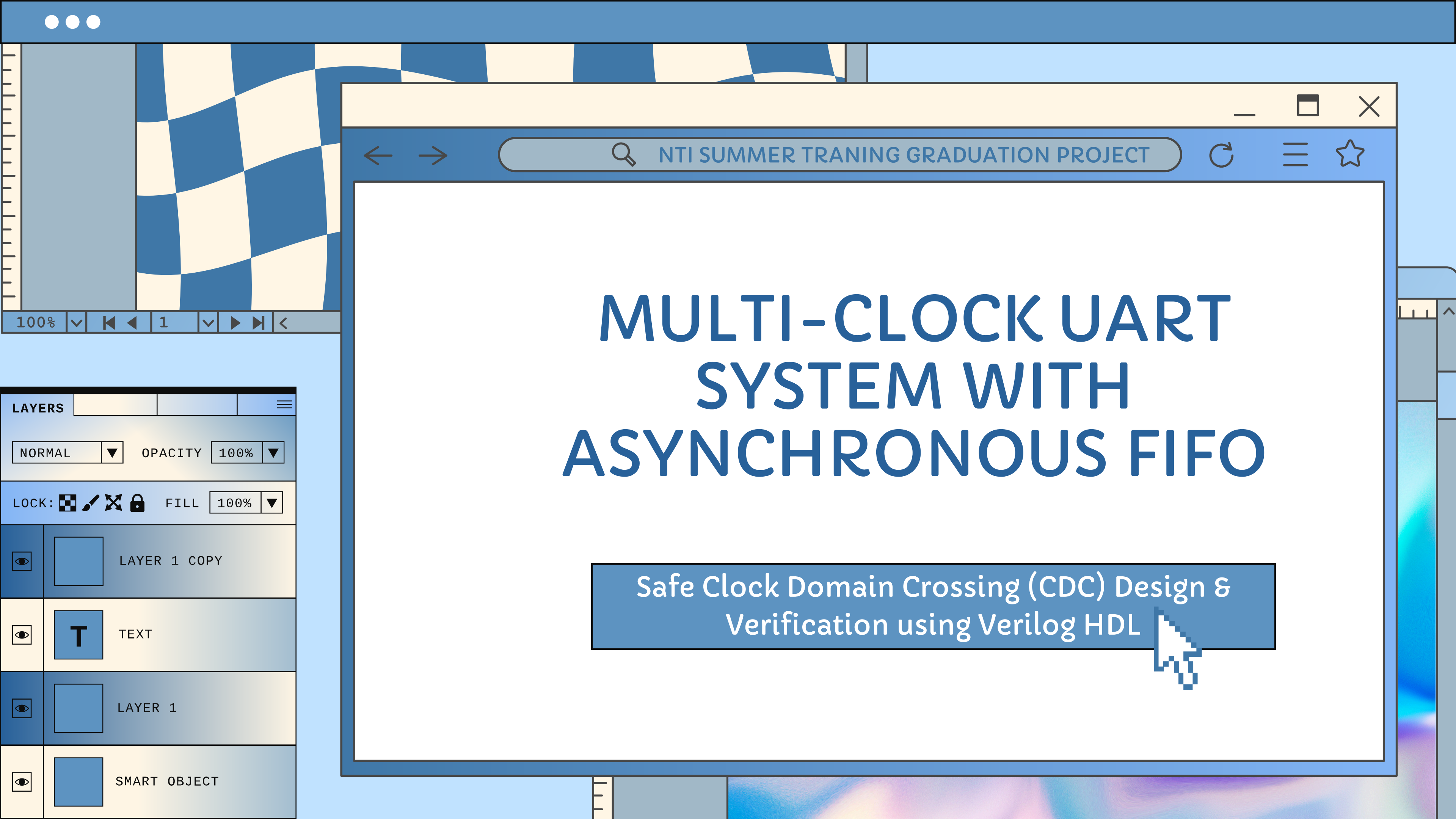




NTI SUMMER TRANING GRADUATION PROJECT

MULTI-CLOCK UART SYSTEM WITH ASYNCHRONOUS FIFO

Safe Clock Domain Crossing (CDC) Design &
Verification using Verilog HDL

LAYERS

NORMAL

▼

OPACITY

100%

▼

LOCK:

FILL

100%

▼

LAYER 1 COPY

T

TEXT

LAYER 1

SMART OBJECT



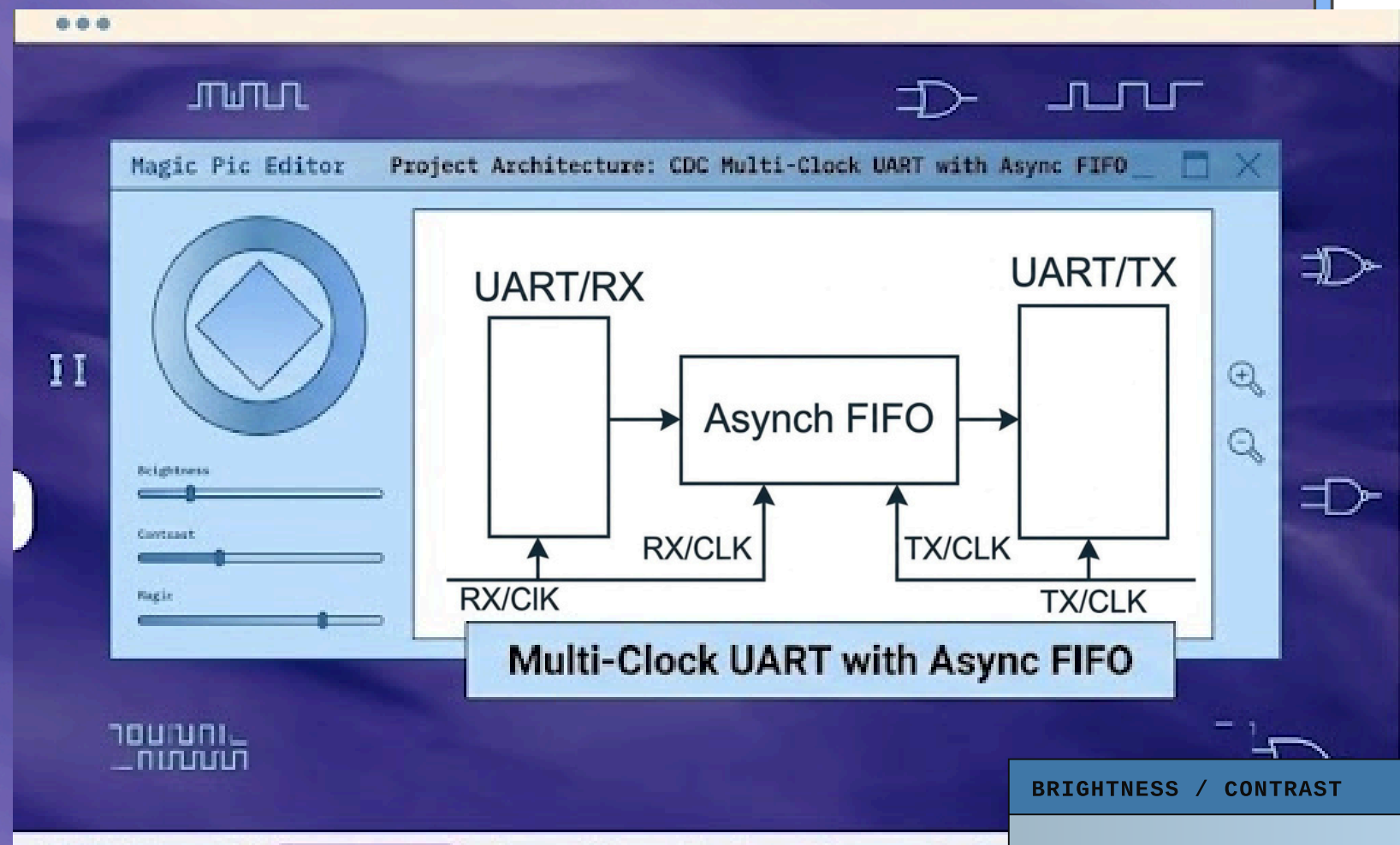
New Tab

<>

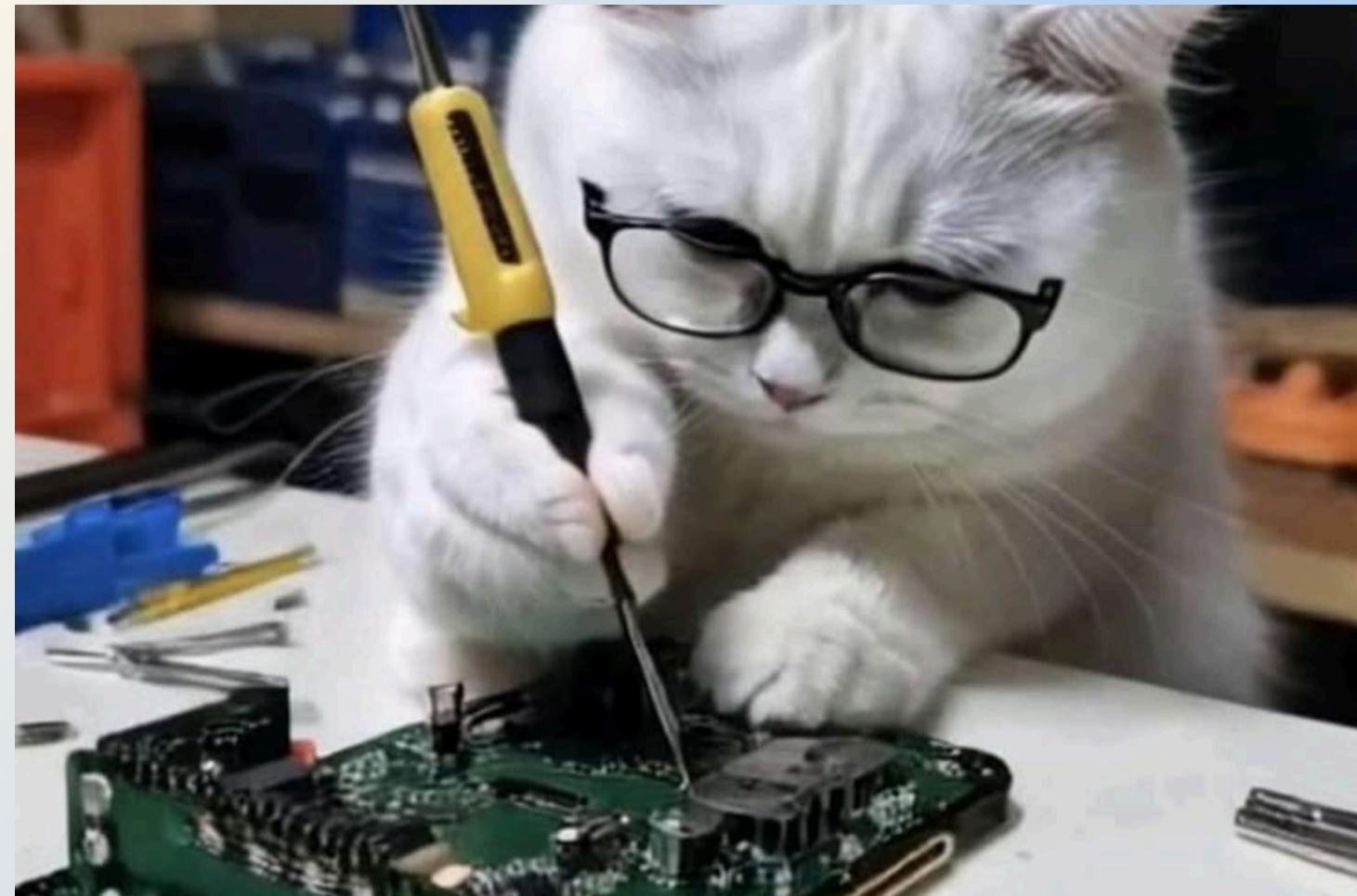
🔍

↺☰

Digital System Design & Verification
Project: Multi-Clock UART System
with Asynchronous FIFO
Date: September 2026



OVERVIEW



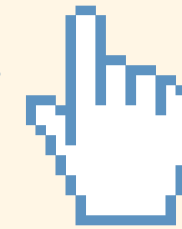
Untitled - TextEdit

File Edit View Help

01

Concept & Project Motivation

- Why we built this project: Solving Clock Domain Crossing (CDC) issues and metastability in multi-clock systems.



02

Block-by-Block RTL Implementation

- Detailed architectural breakdown and Verilog design for UART RX, Async FIFO, and UART TX sub-modules.

03

Integrated System Modeling

- End-to-end multi-clock integration, hardware setup, and data flow modeling.

04

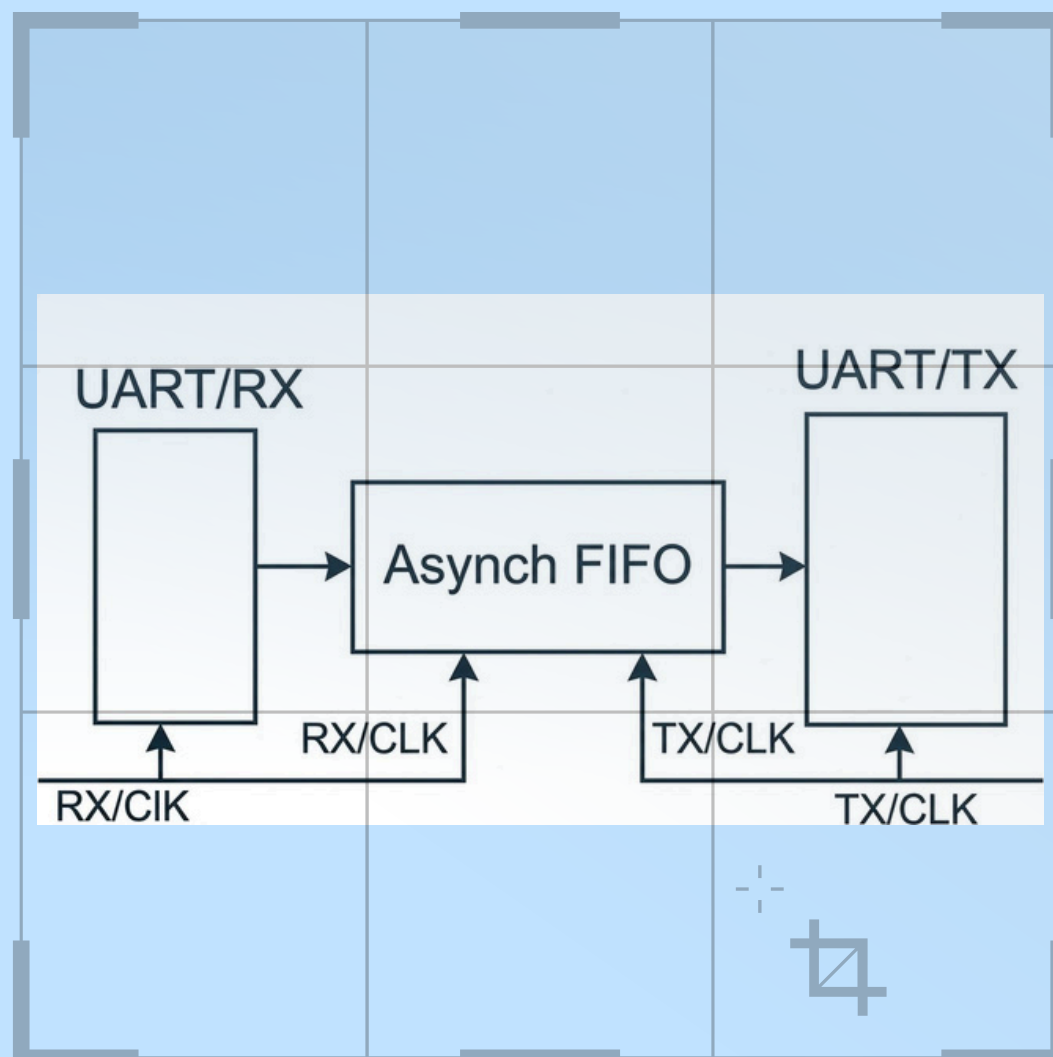
Testbench & Simulation Results

- Self-checking testbench setup, corner-case handling, and ModelSim waveform verification.

CONCEPT & PROJECT MOTIVATION

Page 1

Words: 88



System Integration Challenge

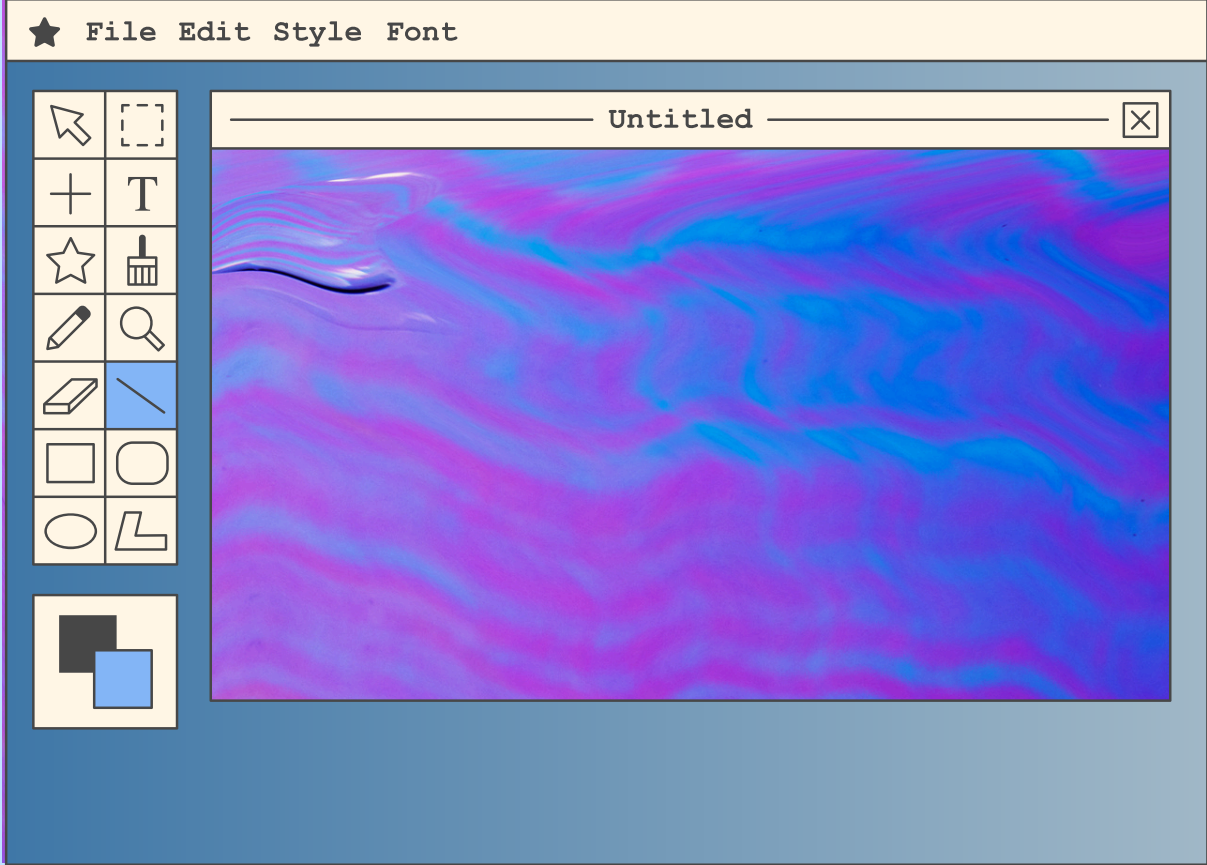
- Heterogeneous SoC Domains
Modern digital systems feature independent, unsynchronized clock frequencies across modules.
- Clock Domain Crossing (CDC)
Direct transfers between fast & slow domains create setup/hold violations, leading to Metastability.
- Rate Mismatch Problem
Fast data bursts from peripherals overload low-speed serial links, causing immediate buffer overflow and frame loss.

Dual-Clock Bridge Solution

- Rate-Matching Engine
Integrates an 8-bit Asynchronous FIFO buffer to absorb fast input bursts (RX_CLK) and meter out slow output streams (TX_CLK).
- CDC Safety Mechanisms
Uses 2-FF Synchronizers and Gray-Coded Pointers to safely pass status flags across clock boundaries.
- Guaranteed Frame Integrity
Achieves zero data corruption and full frame preservation without CPU interventions.

KEY MODULES:

- UART RECEIVER (UART_RX): FEATURES MULTI-RATE OVERSAMPLING (8X, 16X, 32X) AND AUTOMATIC FRAME ERROR DETECTION.
- ASYNCHRONOUS FIFO (ASYNC_FIFO): DUAL-PORT MEMORY BUFFER UTILIZING GRAY-CODE POINTERS AND 2-FLOP SYNCHRONIZERS TO PREVENT METASTABILITY.
- UART TRANSMITTER (UART_TX): SERIALIZER AND FSM-DRIVEN FRAME GENERATOR WITH CONFIGURABLE PARITY GENERATION.



COLOR

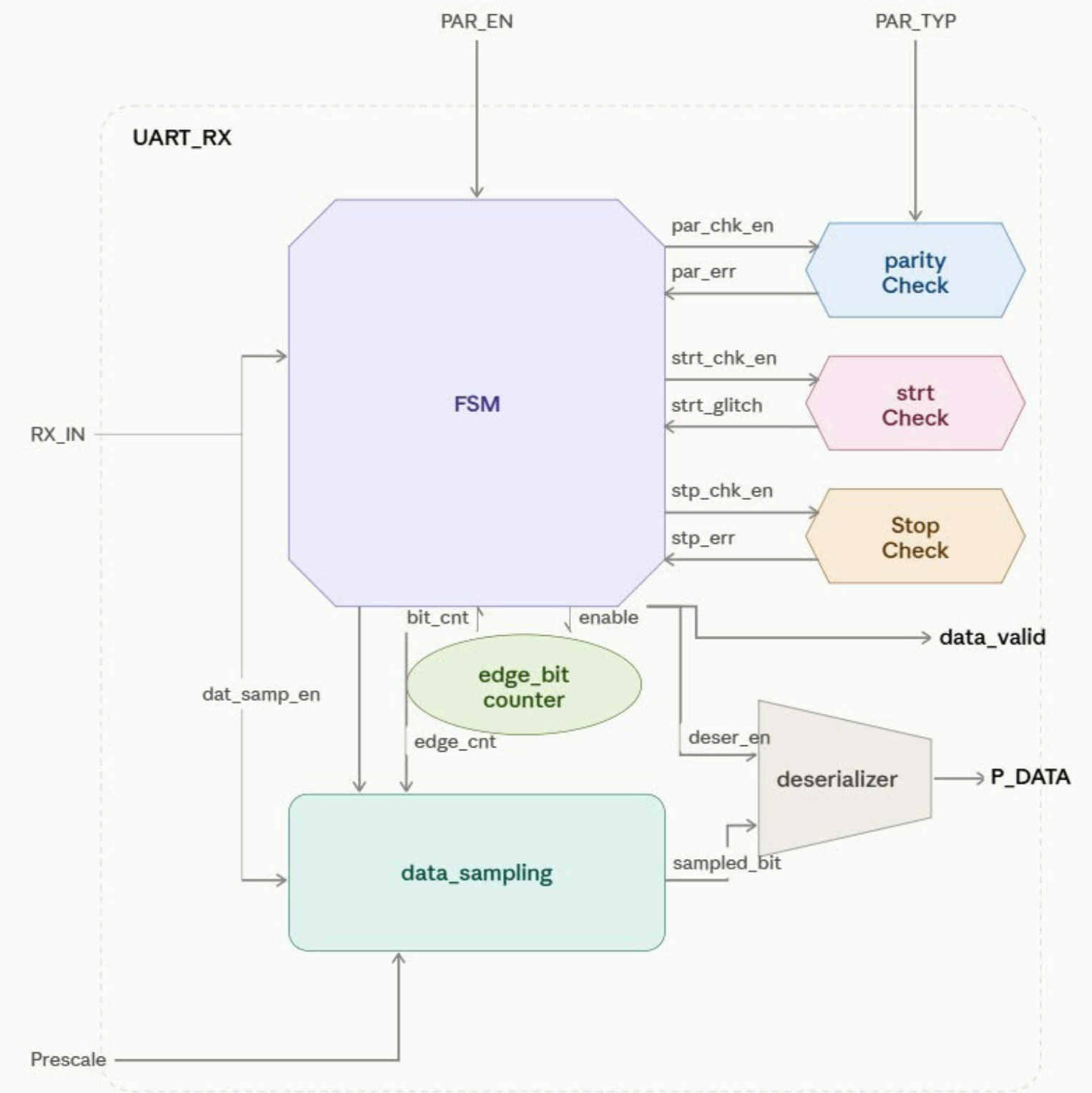
#FFFFFF

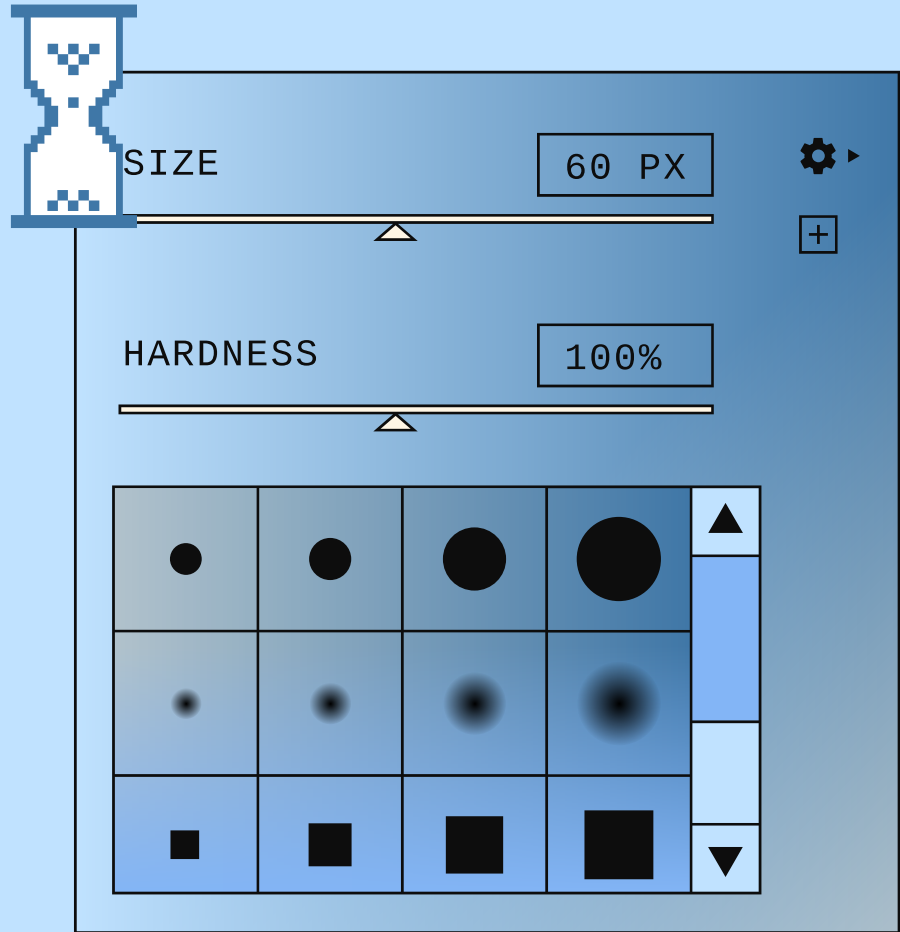
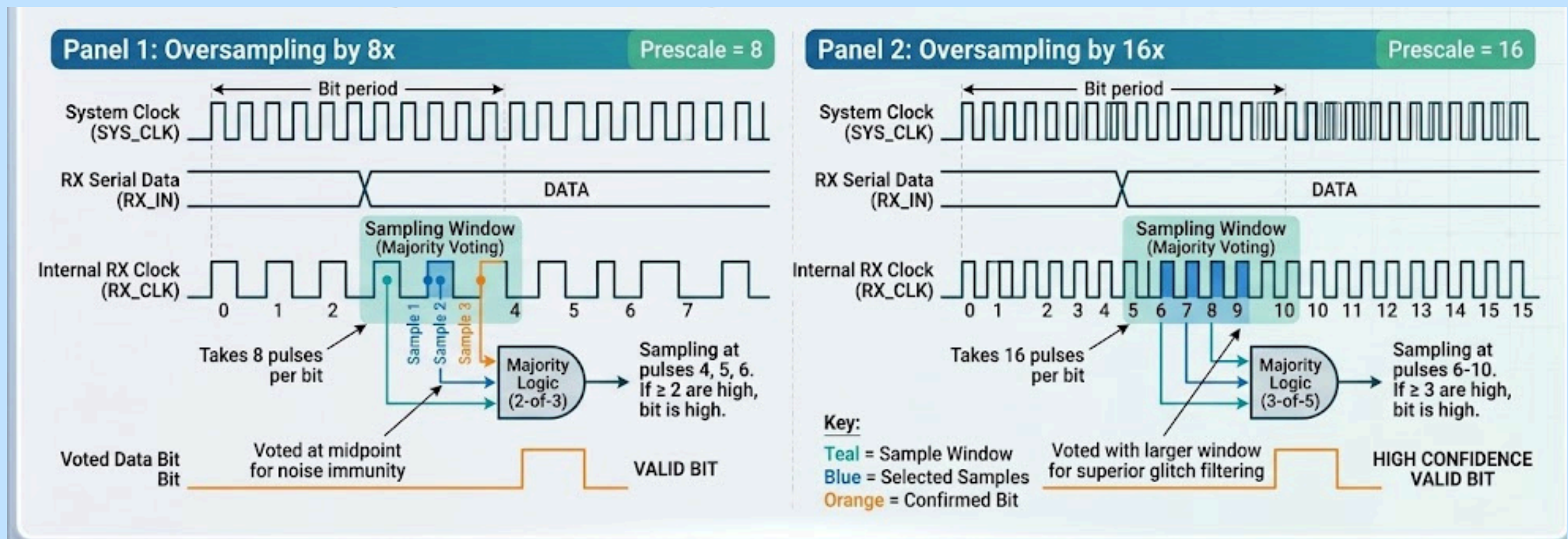
• VERIFICATION STRATEGY: SELF-CHECKING AUTOMATED TESTBENCH ARCHITECTURE FOR FUNCTIONAL AND ERROR-INJECTION TESTING.

UART RECEIVER (UART_RX) STRUCTURAL ARCHITECTURE

File Edit View Help

- **FSM CORE CONTROLLER:** MANAGES STATE EXECUTION, GENERATING ENABLE SIGNALS (DAT_SAMP_EN, DESER_EN, CHECK ENABLES) BASED ON RX_IN STATUS.
- **DATA SAMPLING & BIT COUNTERS:** DATA_SAMPLING PROCESSES OVERSAMPLED BITS WHILE EDGE_BIT COUNTER TRACKS SAMPLE EDGES (EDGE_CNT) AND FRAME BIT POSITIONS (BIT_CNT).
- **DESERIALIZER:** CONVERTS VERIFIED SERIAL BITS (SAMPLED_BIT) INTO 8-BIT PARALLEL BYTE PAYLOAD (P_DATA).
- **HARDWARE VERIFICATION UNITS:**
- **PARITY CHECK:** VALIDATES PARITY ACCORDING TO PAR_TYP CONFIGURATION.
- **STRT CHECK & STOP CHECK:** DETECTS START-BIT GLITCHES AND STOP-BIT FRAMING ERRORS.
- **VALID DATA ASSERTION:** GENERATES DATA_VALID PULSE STRICTLY UPON SUCCESSFUL FRAME VALIDATION WITHOUT ACTIVE ERROR FLAGS.





UART RECEIVER (UART_RX)

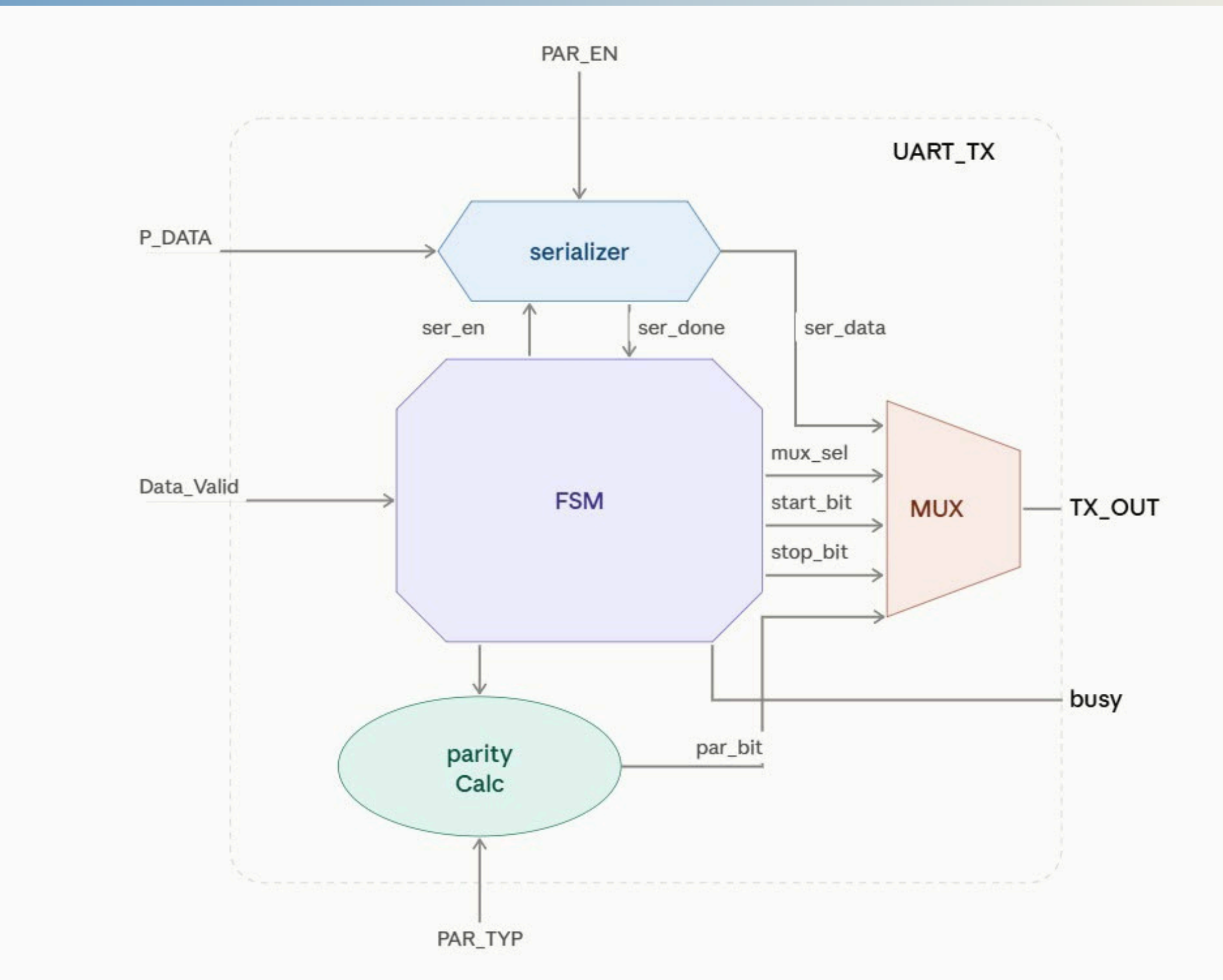
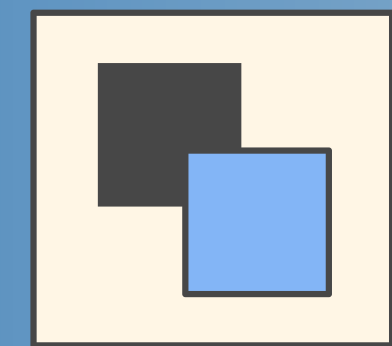
DATA SAMPLING LOGIC

- Prescale-Driven Resolution: Supports dynamic oversampling modes (8x, 16x, 32x) via input Prescale configuration.
- Majority-Voting Technique: Takes 3 samples from the bit midpoint (samples 4, 5, and 6 in 8x mode) to filter out line noise and transient glitches.
- Robust Bit Decision: Drives sampled_bit High if ≥ 2 samples are High, guaranteeing accurate data recovery across asynchronous clock

UART TRANSMITTER (UART_TX) ARCHITECTURE

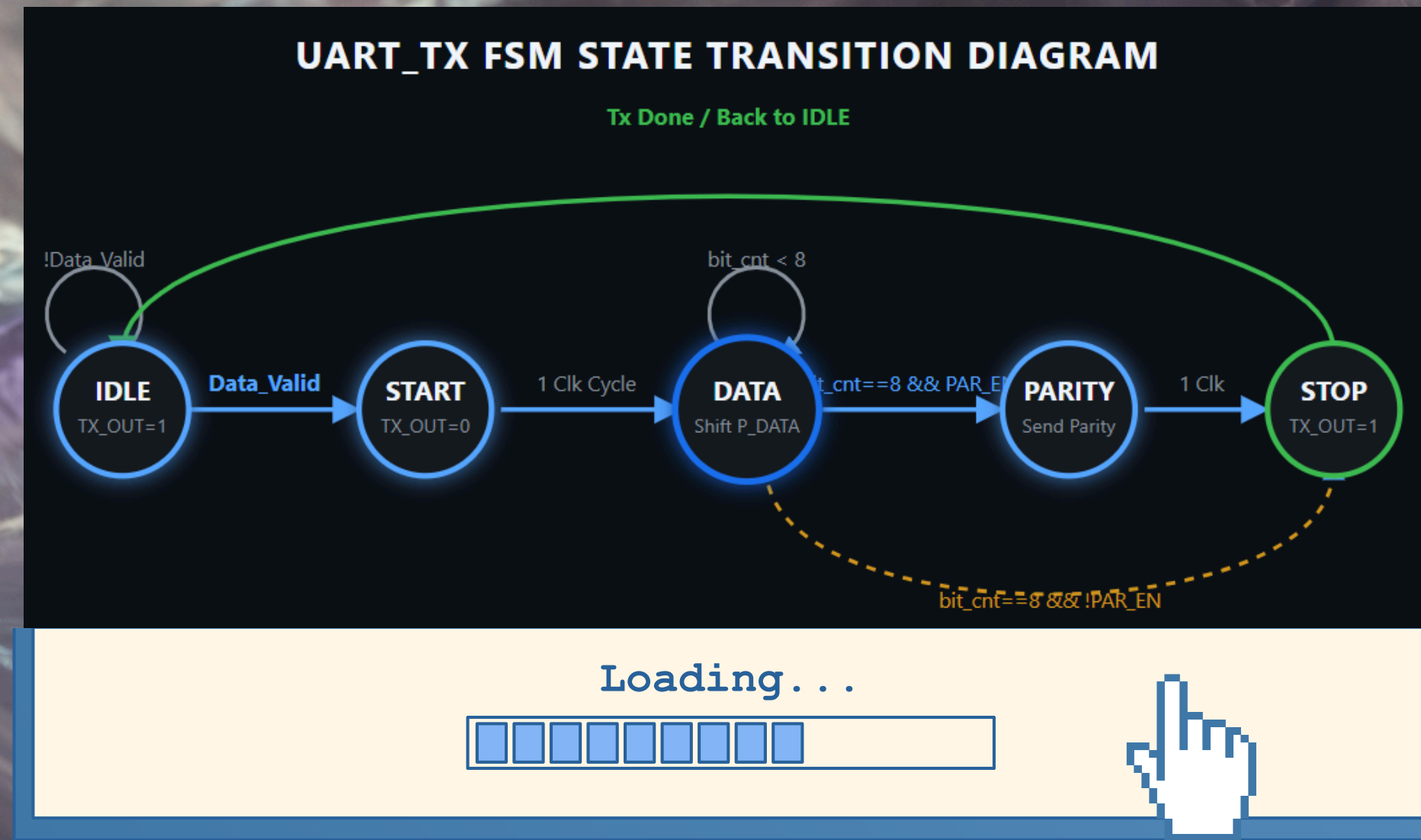
- **Parallel-to-Serial Conversion:** Serializer shifts out the 8-bit parallel payload (P_DATA) LSB-first onto TX_OUT.
- **Dynamic Parity Generation:** Hardware logic calculates odd or even parity on-the-fly based on input frame settings.
- **Multi-Input Multiplexer:** Dynamically routes Start bit, Serialized Data, Parity bit, and Stop bit to the output line.
- **Busy Handshake Flag:** Asserts Busy signal during transmission to block new writes and prevent frame collisions.

🖱️	📐
+	T
★	🖌️
✎	🔍
📄	📏
◻️	◯
◯	📐



File Edit View Help

- IDLE State: Keeps TX_OUT High and monitors incoming Data_Valid trigger pulse.
- START State: Pulls TX_OUT Low for 1 bit-period to signal frame initialization.
- DATA State: Shifts out 8 data bits sequentially synchronized with the baud rate clock.
- PARITY & STOP States: Appends dynamic parity bit (if enabled) and terminates with a High Stop bit.



Document2 - Macrostuff Board

File Edit Format View Help

Cool New Font 12 B I U

UART TRANSMITTER (UART_TX) FSM CONTROL LOGIC

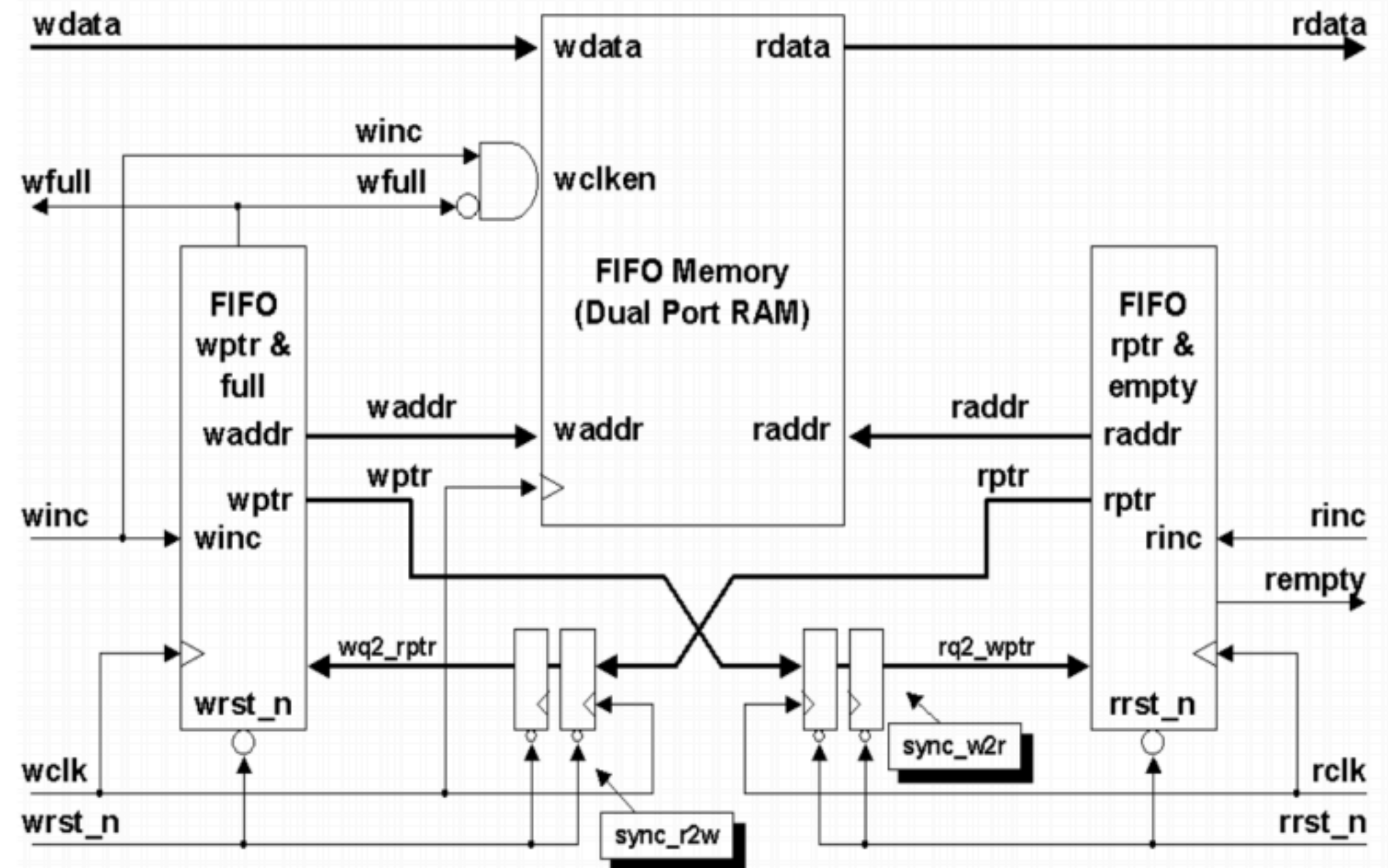
SAVE CHANGES?

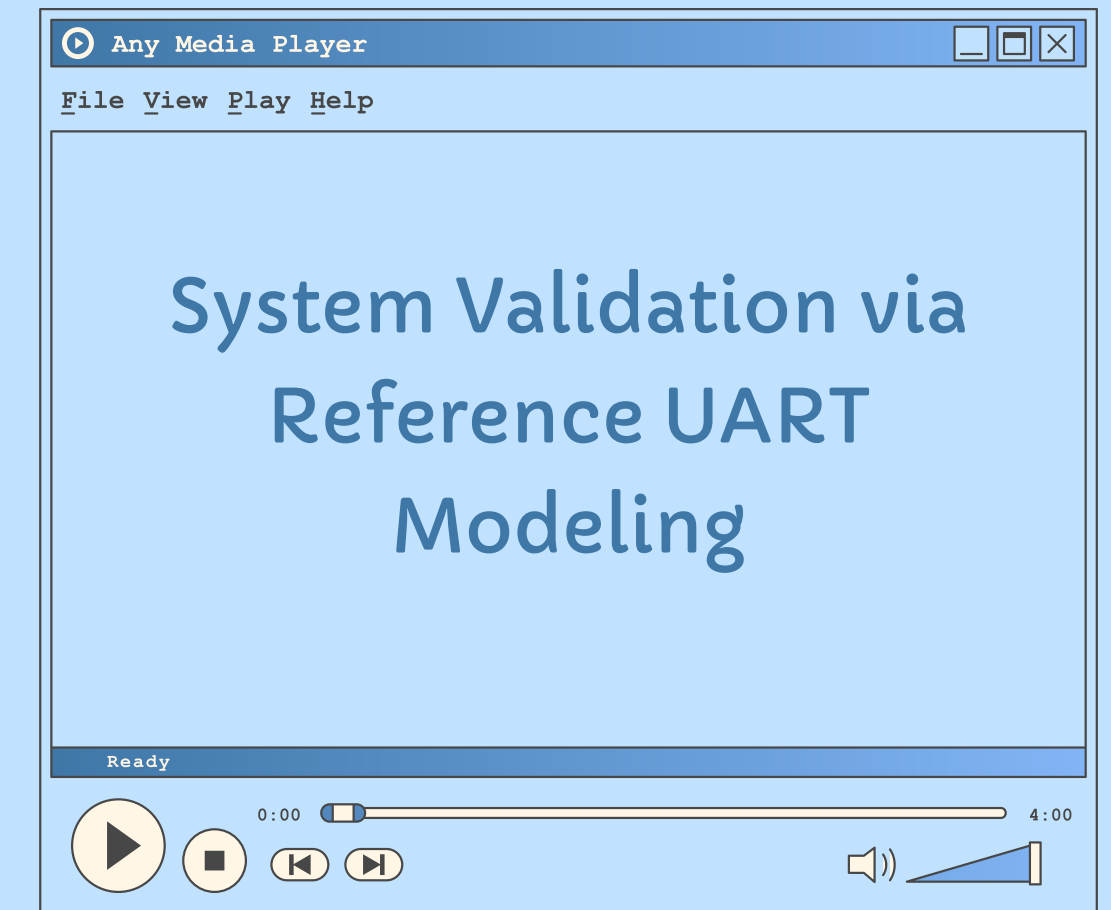
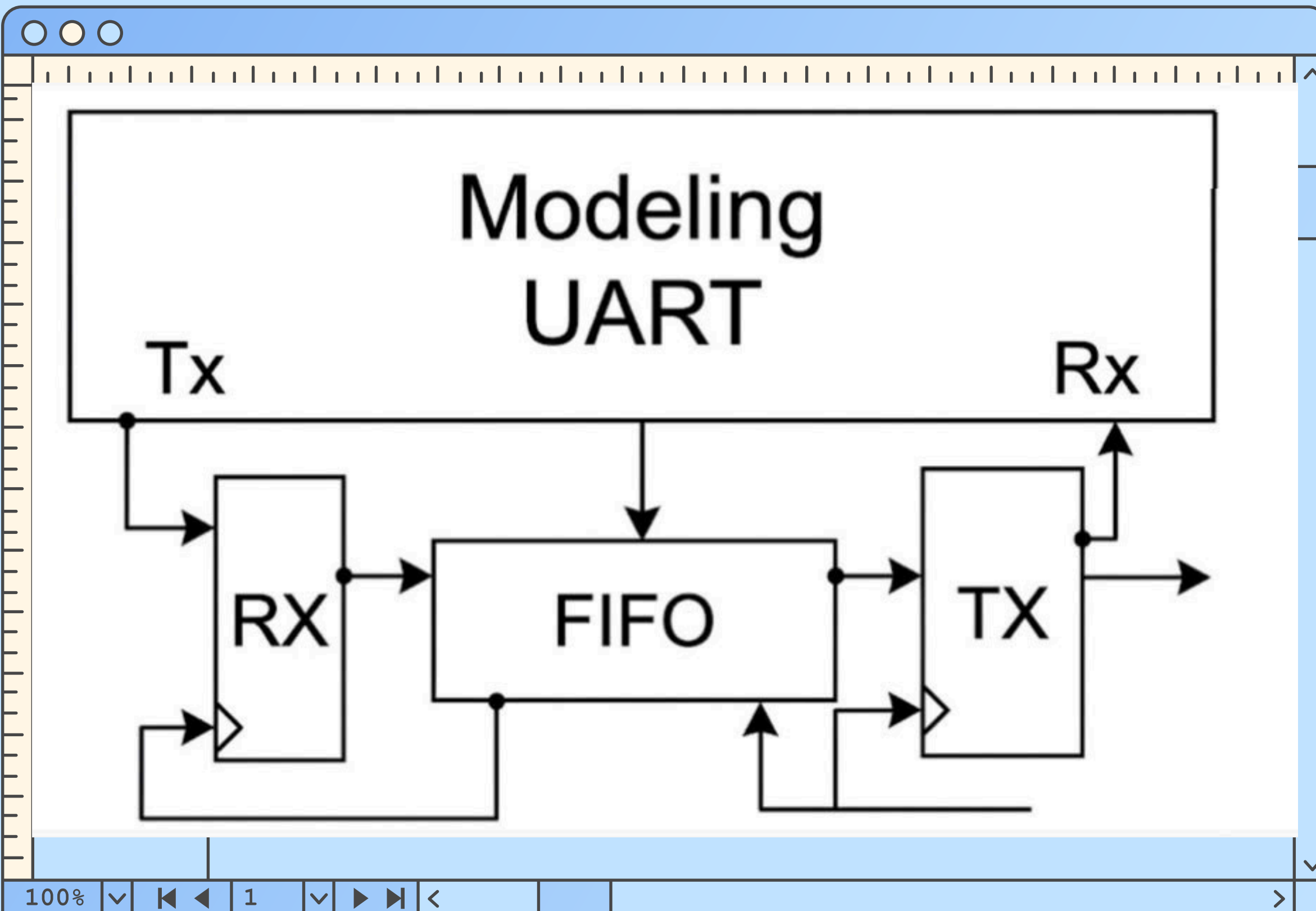
DO YOU WANT TO
YOUR FILE BEFORE

DON'T SAVE

ASYNCHRONOUS FIFO & CDC PROTECTION ARCHITECTURE

- DUAL-PORT RAM CORE: STORES INCOMING BYTES DYNAMICALLY, SEPARATING WRITE DOMAIN (\$WCLK\$) FROM READ DOMAIN (\$RCLK\$).
- GRAY-CODE POINTER ENCODING: CONVERTS BINARY ADDRESS POINTERS INTO GRAY CODE TO GUARANTEE SINGLE-BIT TRANSITIONS DURING DOMAIN CROSSING.
- 2-FLIP-FLOP SYNCHRONIZERS (SYNC_R2W & SYNC_W2R): MITIGATES METASTABILITY BY SAFELY PASSING READ POINTERS TO WRITE DOMAIN AND WRITE POINTERS TO READ DOMAIN.
- TIMING-SAFE FLAG GENERATION: GENERATES CONSERVATIVE WFULL AND REMPTY STATUS FLAGS TO PREVENT BUFFER OVERFLOW AND UNDERFLOW DURING DATA BURSTS.



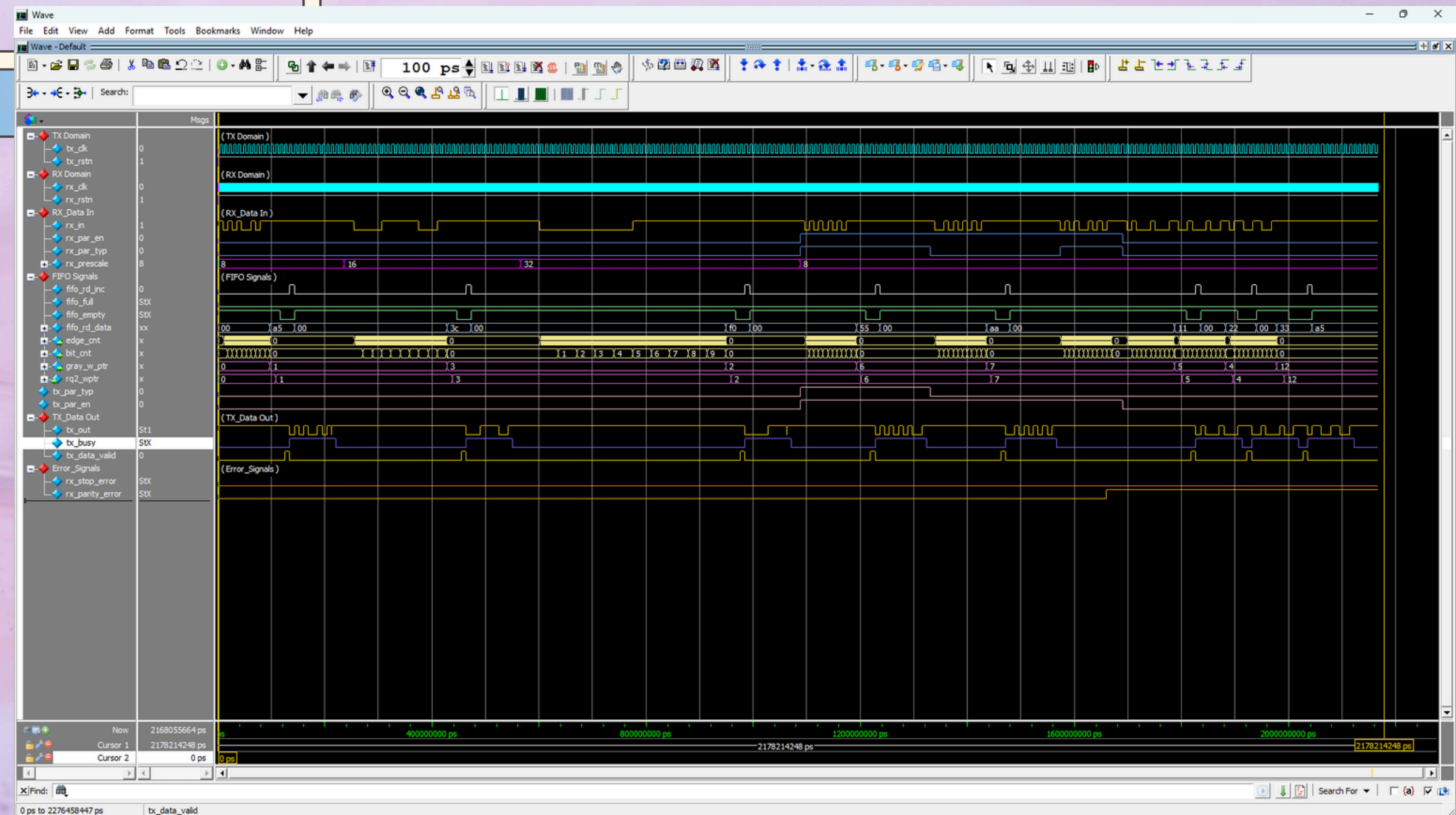


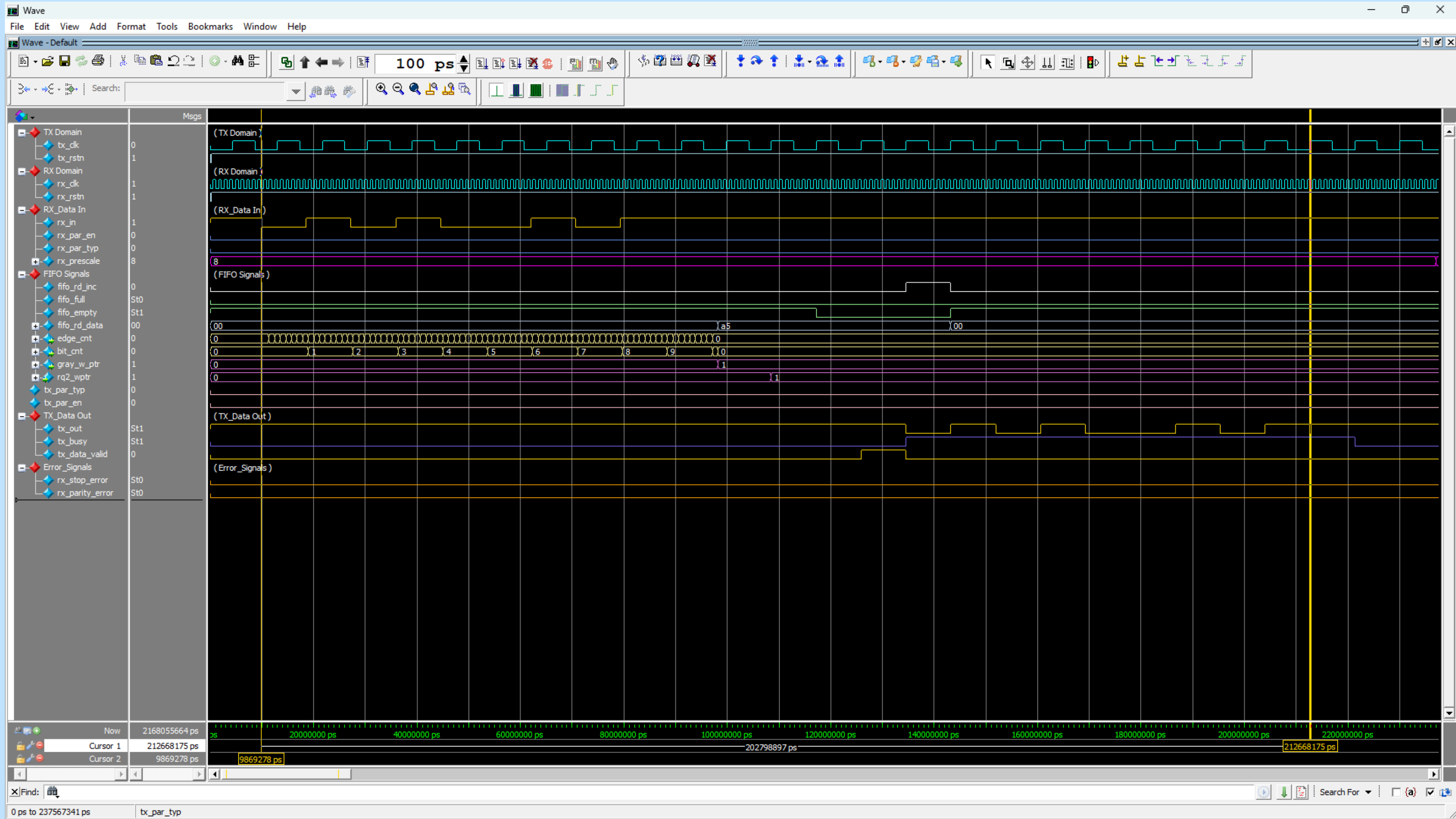
- DATAFLOW: THE REFERENCE TX SENDS DATA TO OUR RX, TRANSFERS IT THROUGH THE FIFO, AND SENDS IT BACK VIA OUR TX TO THE REFERENCE RX.
- OBJECTIVE: COMPARES OUTPUT WITH INPUT TO GUARANTEE STANDARD UART COMPLIANCE, CORRECT TIMING, AND ZERO DATA LOSS.

END-TO-END WAVEFORM & SIMULATION ANALYSIS



img000.jpg





```
STARTING AUTOMATED LOOPBACK VERIFICATION
=====

--- 1. Prescale Sweeps ---
[PASS] Test 1: Transmitted 0xa5 successfully matched 0xa5
[PASS] Test 2: Transmitted 0x3c successfully matched 0x3c
[PASS] Test 3: Transmitted 0xf0 successfully matched 0xf0

--- 2. Parity Hardware Checks ---
[PASS] Test 4: Transmitted 0x55 successfully matched 0x55
[PASS] Test 5: Transmitted 0xaa successfully matched 0xaa

--- 3. Parity Error Injection ---
[TB] Inserting intentional Parity Error...
[PASS] Test 6: rx_parity_error successfully caught intentional parity fault!

--- 4. Back-to-Back Burst Stream ---
[PASS] Test 7: Transmitted 0x11 successfully matched 0x11
[PASS] Test 8: Transmitted 0x22 successfully matched 0x22
[PASS] Test 9: Transmitted 0x33 successfully matched 0x33

=====
TEST SUMMARY REPORT
Total Tests Executed : 9
Total Errors Found   : 0
VERIFICATION STATUS  : ALL PASSED
=====
```



- 100% PASS RATE: ALL 9 AUTOMATED TEST CASES EXECUTED SUCCESSFULLY WITH ZERO DATA MISMATCHES OR ERRORS.
- COMPREHENSIVE TEST COVERAGE:
- PRESCALE SWEEPS: VERIFIED DATA INTEGRITY ACROSS VARYING BAUD-RATE CONFIGURATIONS.
- PARITY & ERROR INJECTION: CONFIRMED CORRECT PARITY LOGIC AND VERIFIED RX_PARITY_ERROR FLAG UNDER INJECTED FAULTS.
- BURST MODE TRANSFER: VALIDATED CONTINUOUS BACK-TO-BACK STREAM HANDLING THROUGH THE ASYNCHRONOUS FIFO.

Synthesis (hierarchy)

- Top-Level Gate-Level Schematic: Synthesized netlist view (via Synopsys Design Vision) showing gate-level mapping of sub-modules (u_uart_rx, u_async_fifo, u_uart_tx).
- Interconnect Integrity: Confirms clean physical interconnects, proper port mapping, and structural hierarchy prior to place-and-route.

watch_this.mp4

Untitled - TextEdit

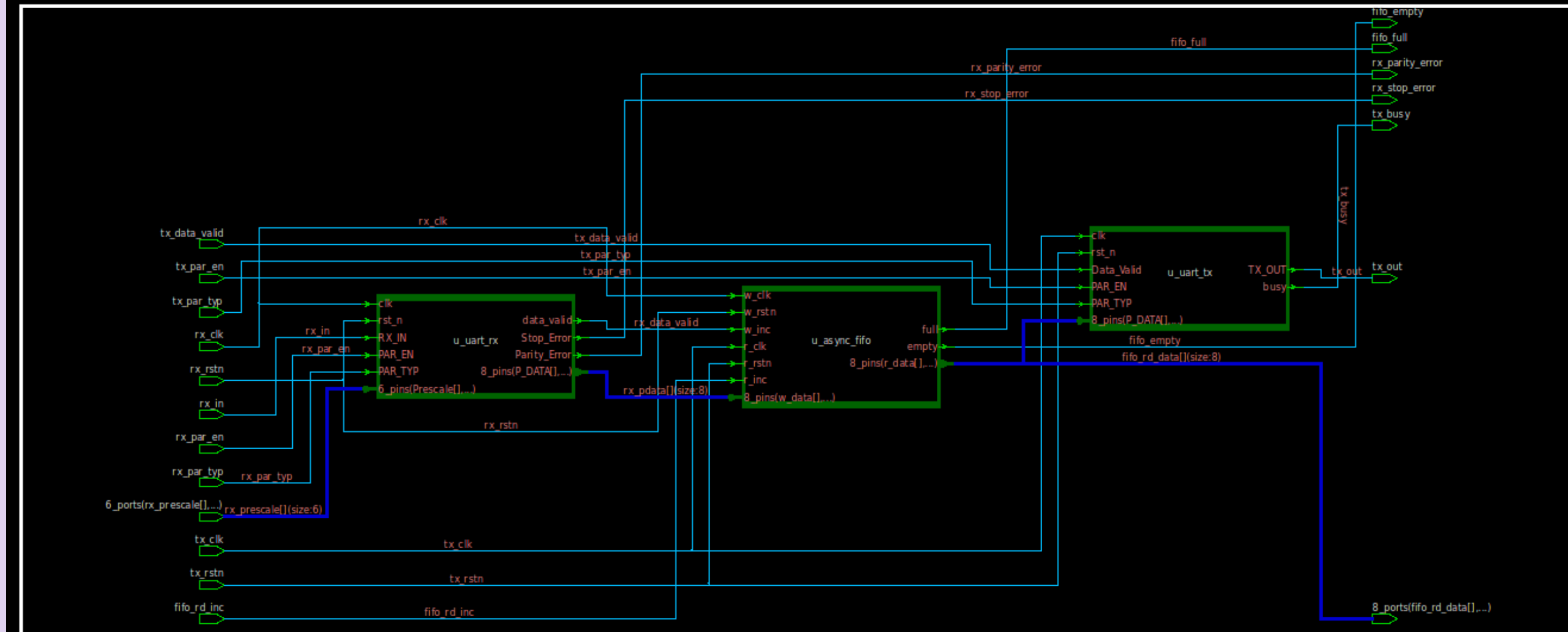
File Edit View Help

Design Vision - TopLevel.1 (system_top)

Design Vision - TopLevel.1 (system_top) - [Schematic.2]

Highlight List Hierarchy Design Attributes Schematic Timing Test Power AnalyzeRTL Window Help

system_top



Hier.1

[1=0]

Schematic.2

select (Hold Ctrl to add, Shift to remove)

Design

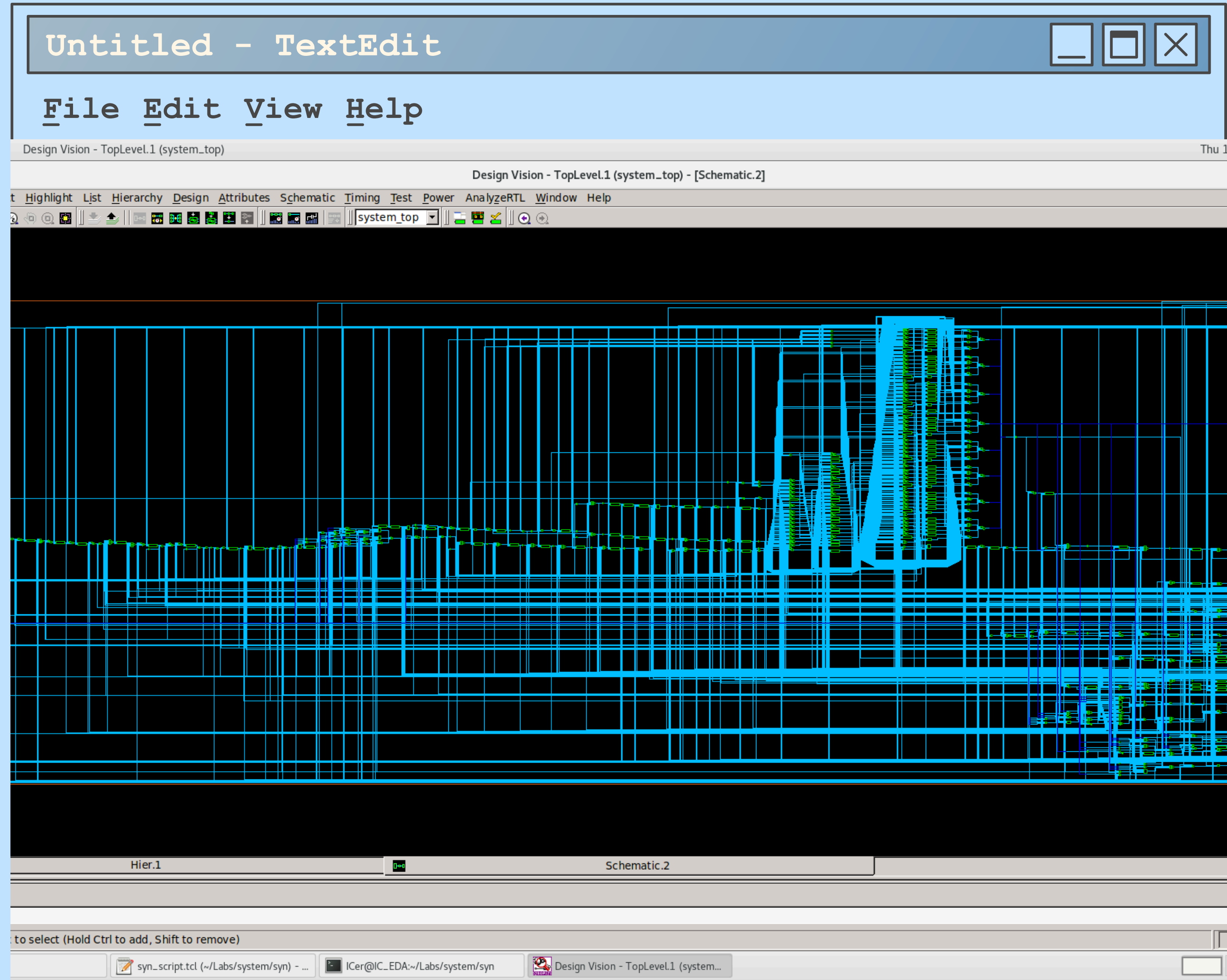
Synthesis (flattened)

Video Viewer

1080px (W)1920px (H)

- Flattened Gate-Level Netlist:
Comprehensive physical mapping with all design hierarchy resolved into a single level for optimized logic synthesis.
- Physical Verification Ready: Final structural view of individual gates and connections, prepared for backend place-and-route processes.

watch_this.mp4



Thank you

